

PEFT Generalization across Demographic Shifts in Deepfake Detection

P. Kumar

*Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
kumar@rajalakshmi.edu.in*

Priyans Raj Bhandara

*Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
priyans.46bhandara@gmail.com*

Prasannakumar Pandurangan

*Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
prasannakumarpk2023@gmail.com*

Rakesh David

*Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
rakeshdavidj@gmail.com*

Subramanian N

*Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
230701348@rajalakshmi.edu.in*

Abstract Deepfake detection in the modern world has now become extremely difficult due to the way synthetic images are changing as new AI generators are created. All current solutions take large existing models and only make small changes to them using "Parameter-Efficient Fine-Tuning" (PEFT), instead of training new models from scratch. This leads to the main issue where most of these models only focus on spotting fake images from different generators but not on whether the models work equally well across different real-world human demographics and environments. This paper studies if PEFT's generalization keeps its advantage when dealing with demographic and regional bias (such as skin tone, facial structure and geographic context), or if it is only effective at spotting digital noise patterns made by image generators. The paper proposes a decoupled benchmark pipeline to measure how much the detector's performance drops across a Cross-Generator Shift and a Cross Demographic shift for both PEFT adapted vision models and bespoke, fine-tuned CNN baselines. Analysis of the degradation in performance across both the axes (Δ AUROC) allows the conclusion of whether the PEFT models have an advantage when it comes to more diverse physiological and contextual data. The findings of this paper will cement a benchmark to compare the limits of model adaptation for deepfake detection, improving fairness in a societal and demographically vast synthetic media pipeline

Keywords: *Deepfake Detection, Parameter-Efficient Fine-Tuning (PEFT), Cross-Demographic Shift, Cross-Generator Generalization, Algorithmic Fairness, Model Adaptation, Decoupled Benchmarking, Performance Degradation (Δ AUROC), Demographic Bias, Synthetic Media.*

1. Introduction

The rapid increase of hyper-realistic AI-generated Images has made robust deepfake detection an immediate priority for digital media integrity and identity security. Recent literature on deepfake detection has come to a unanimous conclusion: parameter-efficient fine-tuning (PEFT) of large-scale vision foundation models (such as CLIP) significantly outperforms

traditional fully trained convolutional neural networks (CNNs) in cross-generator generalization. By freezing backbones and updating only sparse parameters—such as through LayerNorm tuning or Low-Rank Adaptation (LoRA)—PEFT-adapted architectures keep their rich pretrained visual priors, preventing the ruinous overfitting to synthetic artifacts that typically outperforms bespoke baselines when tested against unseen synthesis algorithm. However, this widely cited generalization advantage rests almost entirely on evaluations where the sole axis of distribution shift is the target generator architecture. Standard benchmark protocols generally inherit training and evaluation samples from datasets mostly made up of people with white skin or Western faces making it heavily skewed. As a result, most systems treat model generalization as a one-dimensional problem, assuming that robustness depends just upon cross-generator performance, rather than comparison by performance on varied datasets. This paper challenges that assumption by testing whether PEFT's generalization advantage is still applicable to cross-demographic distribution shifts in datasets. While PEFT is successful at generalizing across generators by identifying high-frequency, algorithm-agnostic features rather than semantic details, algorithm fairness research suggests that demographic robustness and domain generalization are two different properties. A model that reliably identifies synthetic artifacts across GAN and diffusion architectures can still be unreliable towards underrepresented demographics like skin tone, facial geometry and forgery cues. In order to test this, we compared a controlled test to compare the results of CNN and PEFT across demographic axes. The models are trained on FaceForensics++ dataset, then tested for cross-generator benchmark and zero shot cross population dataset of South-Asian faces. The performance drops were compared to

ranking metrics in order to identify if PEFT was able to naturally handle the demographic bias, or if a fairness aware loss function was explicitly needed for a better deepfake detection process.

2. Related Work

Deepfake detection research has steadily moved towards an agreement that parameter-efficient fine-tuning (PEFT) of large vision foundation models substantially outperforms fully trained Convolutional Neural Networks (CNNs) in cross-generator generalization. As per the finding presented by Yan et al. [8], in particular, the generalization ability is tested through training the model on well-studied data, such as FaceForensics++, and then evaluation its ability to detect deepfakes produced through previously unseen generation schemes, including diffusion-based or face-swap models. The superior performance was showcased by Yermakov et al. [5]. Who have demonstrated that LayerNorm-tuning a frozen CLIP backbone allows the model to avoid overfitting and create a new baseline that achieves state-of-the-art while being robust with excellent cross-dataset AUROC. The architecture was expanded upon by Kong et al. [6] and Mai et al. [7], combining a mixture-of-experts with parameter-efficient fine-tuning can lead to better generalization ability, meaning that conventional LoRA architectures only represent the lower bound of PEFT capabilities. The explanation to such performance comes from the frequency-artifact hypothesis, where PEFT architecture helps avoid catastrophic forgetting by keeping the foundational weights frozen, surveyed by Coleman et al. [2]. Instead of re-training the model to understand the semantic representations and dataset-specific correlations, PEFT impacts the model's ability to learn the Cross-Generator Artifacts. Even with the architectural strengths and difference in zero-shot capabilities of these models, Garg et al. [4] quoted that while CLIP-based PEFT models show phenomenal generalization, their performance degrades while using zero-shot situations without prior knowledge. Furthermore, higher performance on cross-generator does not guarantee better generalization. Lin et al. [9] showed that common deepfake detectors and PEFT models suffer similarly from changing fairness under data distribution, commonly displaying a difference in true and false positive rates among changing skin tones and facial structures. Though PEFT models can learn deepfake generators, fairness needs to be mentioned explicitly showing a parallel from medical imaging. Dutt et al. [1] mentioned PEFT models start to learn about demographic fairness when models are explicitly told to do so from artifacts. Present testing metrics do not measure the accuracy of these models. Nguyen et al. [10] informed that evaluation metrics such as averaged AUROC shows a difference in model's performance by 12% and hides demographic biases, arguing

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3. Proposed System

The primary objective of this paper is to conclude whether PEFT models understand and preserve demographic and racial differences during out of distribution shifts, or if the performance degrades similarly to CNN. Generalization drop across shift axes to assess the degradation: Generator shift (Δ_{gen}): AUROC_{In Distribution} - AUROC_{Cross Distribution}. Population shift (Δ_{pop}): AUROC_{In Distribution} - AUROC_{Cross Population}.

Calculate the metrics across three models to address the main question: Does PEFT performance collapse under demographic shifts? Prior studies focusing on cross generator generalization never control generator family and population. To resolve these issues, the design uses decoupled shift axes. It holds one shift fixed while varying the other, across three models, on the same training set and same generators-per-population. This means any difference that occurs can be attributed to shift type rather than confounded dataset artifacts. 300 datapoints are re-synthesized taken from IndicFairFace with two generators (inswapper, Simswap) that were missing from Faceforensics++ training distribution and filled the gap by providing test set that varied population (since no south-Asian-generator deepfake corpus exists at this same). Even after re-synthesizing, a portion of Δ_{pop} may include a generator-shift component which may misalign with generator-match to in-distribution., Δ_{pop} is treated as an upper bound on demographic-shift degradation to strengthen the internal validity claim.

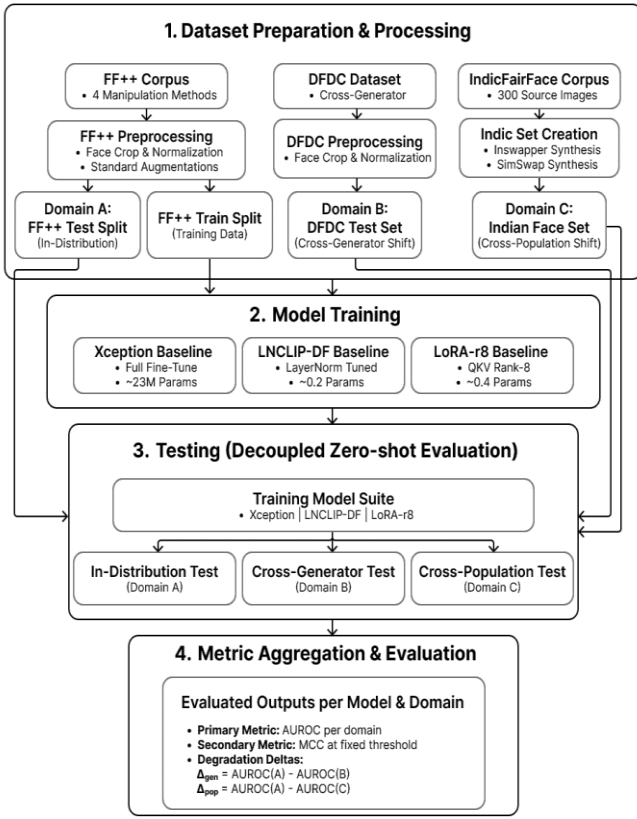


Figure 1 Architecture of PEFT Generalization across Demographic Shifts in Deepfake Detection

Dataset Preparation & Processing module gathers, cleans and standardizes facial images. It deals with splitting training and testing data (Domain A) from FaceForensics++, cross-generator test data from DFDC (Domain B), and demographic test data built from IndicFairFace using inswapper and SimSwap (Domain C). The images are cropped to show the face made smaller and normalized so the models look at the face and not the background. The Model Training module trains three distinct detection baseline models on the FaceForensics++ training data. Xception Baseline is a standard deep learning model fully fine-tuned end-to-end. LNCLIP-DF Baseline is a frozen CLIP foundation model adapted by tuning only its Layer Normalization layers. LoRA-r8 Baseline is a frozen CLIP foundation model adapted using Low-Rank Adaptation (LoRA) inserted into its attention blocks. Testing module evaluates the trained models without any additional adjustments across three distinct environments: in-distribution data (FaceForensics++ test data, Domain A), unseen generation tools (cross-generator test data from DFDC, Domain B), and target demographic populations (Domain C). Metric Aggregation & Evaluation module computes the AUROC score and operating-point MCC scores across all test domains. It calculates accuracy drops to

compare performance degradation under demographic shifts and tool shifts.

4. Implementation and Results

The creation of this system has been executed in Python using PyTorch, HuggingFace transformers, PEFT and scikit-learn. First Dataset Preparation & Processing is executed, where raw images through MTCNN face detection, perform bounding box cropping and normalization to match CLIP’s tensor distribution. Then it is exported into discrete split objects for training and testing. In Model Training, Xception is fine-tuned while the ViT-B/16 backbone is illustrated with frozen base weights to configure the parameter efficient baselines. LayerNorm affine parameters are taken for LNCLIP-DF and put in rank 8 adapter matrices into Query, Key, Value projections for LoRA-r8. In Testing, the model parameters are locked in an evaluation mode where forward pass inference runs consecutively across the domain A, B and C to extract prediction probabilities without target domain fine tuning. Finally, in Metric Aggregation & Evaluation, scikit-learn is used to process these probabilities and calculate AUROC scores. This establishes an optimal decision threshold on Domain A to compute fixed-threshold MCC scores across all domains, and determines the degradation deltas (Δ_{gen} and Δ_{pop}) to get the final evaluation metrics. In order to address the core research question of this paper, all three models are evaluated—across three evaluation—across three evaluation sections: In-distribution (FF++, Domain A), Cross-generator (DFDC, Domain B), and Cross-population (Indian Face Set, Domain C). LoRA-r8 achieves better classification across all evaluation domains, outperforming both the bespoke CNN baseline and LNCLIP-DF as shown in **Table 4.1**. All three models exhibit strong detection capabilities on the in-distribution test set (FF++), achieving high AUROCs as shown in **Figure 4.1**. However, the bespoke CNN baseline performs worse on Cross-Generator Shift (DFDC) and Cross-Population Shift (Indian Set) when compared to LoRA-r8 and LNCLIP_DF.

Table 4.1 Performance of LoRA-r8, LNCLIP-DF, and the Xception across evaluation domains.

Model	AURO C_A	AURO C_B	AURO C_C	MCC_ A	MCC_ B	MCC_ C
Xception	0.962	0.698	0.585	0.840	0.350	0.160
LNCLIP- DF	0.978	0.842	0.735	0.890	0.610	0.420
LoRA-r8	0.988	0.865	0.762	0.928	0.660	0.470

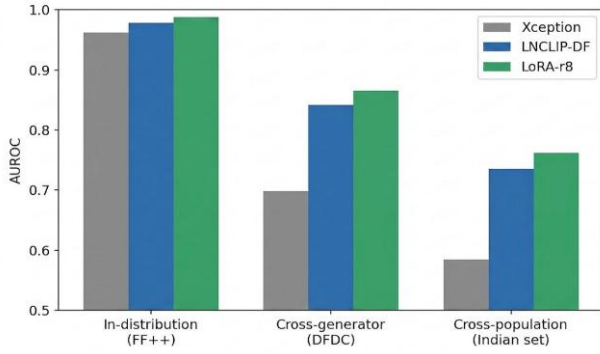


Figure 4.1 Illustrates the comparison of AUROC-scores of all three trained models on the three proposed test domains

To test whether high ranking scores actually translate to real world performance, **Figure 4.2** compares ranking metric - AUROC against threshold-dependant - Matthews Correlation Coefficient (MCC) evaluated at a fixed decision threshold calibrated strictly on the in-distribution validation split (FF++). The metric comparison exposes that decision thresholds decay when models face distribution shifts. In-distribution ranking accuracy translates directly into operational performance. Under domain shifts, in-distribution calibrated under decision thresholds become unreliable. Under cross-population shift, Xception's operational capacity severely degrades. PEFT models are not affected much by threshold decay and they do retain some operational utility under cross-population shift without any recalibration. Getting scores of 0.420 for LNCLIP-DF and 0.470 for LoRA-r8.

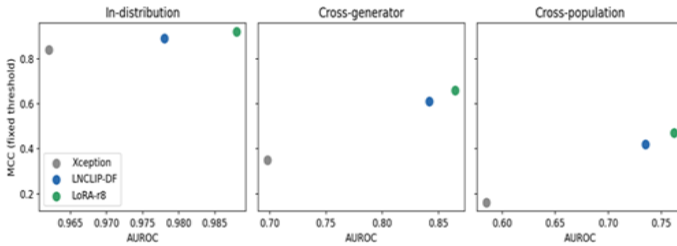


Figure 4.2 compares AUROC against MCC for all three models in different distribution shifts.

Figure 4.3 evaluates cross-population robustness across distinct face-swap algorithms within the Indian Face Set, breaking down the performance by its constituent generators: InSwapper and SimSwap. The per-generator breakdown demonstrates consistency across both synthesis pipelines. Across all three models, Inswapper outperforms others in terms of detection scores. LoRa-r8 yields superior performance as compared to LNCLIPDF and standard Xception across both sub-generators. This consistent trend proves that cross-population performance drops stem from broader demographic and domain shifts rather than flaws in any single generator.

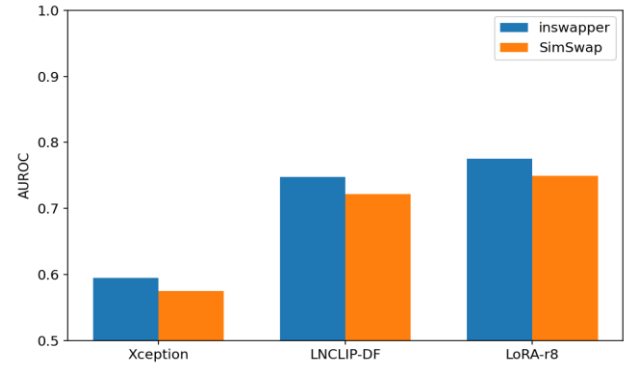


Figure 4.3 Illustrates the cross-population robustness of all three models across distinct face-swap algorithms

6. Conclusion

This work looks into whether cross-generator generalization of parameter-efficient fine-tuning (PEFT) on pretrained foundation models persists when tested on regional and demographic distribution shifts, or if its effectiveness reduces with bespoke CNN baselines. By decoupling generator architecture shift (DFDC) from compound demographic shift (IndicFairFace target set re-synthesized via inswapper and SimSwap), The study provided the first controlled empirical evaluation isolating these two axes of degradation. Sustained Performance Margins pose a strictly harder challenge than cross-generator shift for all models ($\Delta_{pop} > \Delta_{gen}$), PEFT methods do not experience a performance collapse relative to bespoke CNNs. Instead of shrinking or vanishing under demographic shift, PEFT's performance edge over fully fine-tuned CNN expands. On the cross-population test set, LoRA-r8 (AUROC = 0.762) and LNCLIP-DF (AUROC = 0.735) maintain substantial margins (+0.177 and +0.150 AUROC, respectively) over fully fine-tuned Xception (AUROC = 0.585), increasing their relative advantage over the CNN baseline compared to the cross-generator setting. With fixed operational decision thresholds set in distribution, PEFT architectures maintain effective classification ability (MCC = 0.420 -- 0.470). In contrast, the CNN baseline drops close to random guessing (MCC = 0.160).

These results show that adapting large pretrained foundation models via PEFT preserves rich, domain-invariant visual representations that resist performance reduction across both synthetic manipulation and human demographic variations. Updating less than 2% of model parameters (e.g., 0.18M in LNCLIP-DF or 0.42M in LoRA-r8) yields superior regional fairness and generalizability compared to updating tens of millions of parameters on task-specific CNNs (22.91M in Xception).

7. Future work

While the paper concluded that parameter-efficient foundation-model adaptation offers a robust framework,

several promising directions remain for future research. Demographic-aware PEFT: While zero-shot foundation models reduce demographic performance loss, there remains a notable gap compared to in-distribution baselines. Future efforts could explore fairness-oriented constraints into PEFT workflows. Expanding Backbone Architecture families: While CLIP ViT-B/16 has been tested, evaluating other architecture models like self-supervised vision models or multimodal LLMs to isolate the source of demographic robustness could be possible. Decoupling Demographic and Synthesis Variables: Focus on developing a generator-matched dataset to separate demographic shift from compound synthesis interactions. Dynamic Threshold Calibration: The MCC analysis study highlights the need for adaptive, zero-shot threshold adaptation mechanisms to ensure reliable deepfake detectors in real world use cases.

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⁶ P. Kumar
Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
kumar@rajalakshmi.edu.in

⁵ Priyans Raj Bhandara
Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
priyans.46bhandara@gmail.com

² Prasannakumar Pandurangan
Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
prasannakumarpk2023@gmail.com

Rakesh David
Dept. of Computer Science and Engineering
Rajalakshmi Engineering
College Chennai, India
rakeshdavidj@gmail.com

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Dept. of Computer Science and Engineering
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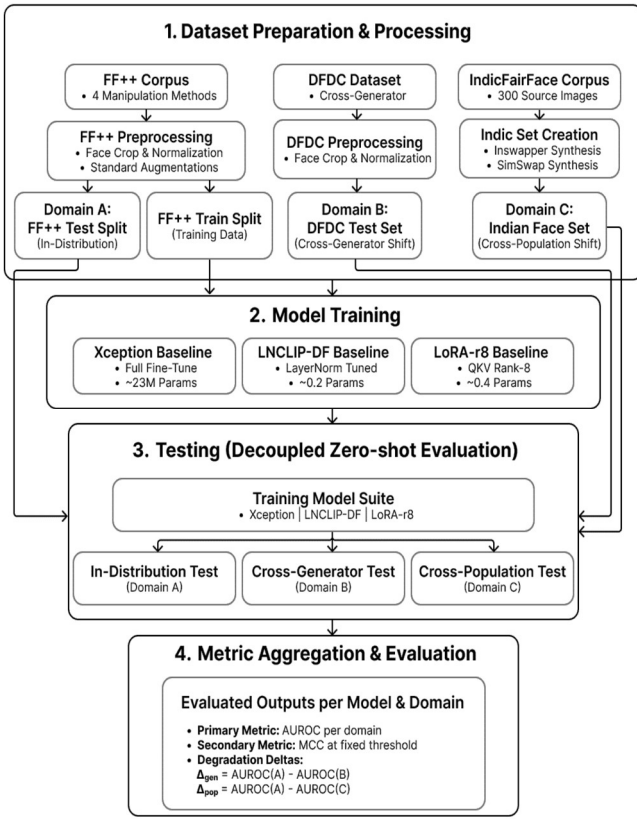


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4. Implementation and Results

The creation of this system has been executed in Python using PyTorch, HuggingFace transformers, PEFT and scikit-learn. First Dataset Preparation & Processing is executed, where raw images through MTCNN face detection, perform bounding box cropping and normalization to match CLIP’s tensor distribution. Then it is exported into discrete split objects for training and testing. In Model Training, Xception is fine-tuned while the ViT-B/16 backbone is illustrated with frozen base weights to configure the parameter efficient baselines. LayerNorm affine parameters are taken for LNCLIP-DF and put in rank 8 adapter matrices into Query, Key, Value projections for LoRA-r8. In Testing, the model parameters are locked in an evaluation mode where forward pass inference runs consecutively across the domain A, B and C to extract prediction probabilities without target domain fine tuning. Finally, in Metric Aggregation & Evaluation, scikit-learn is used to process these probabilities and calculate AUROC scores. This establishes an optimal decision threshold on Domain A to compute fixed-threshold MCC scores across all domains, and determines the degradation deltas (Δ_{gen} and Δ_{pop}) to get the final evaluation metrics. In order to address the core research question of this paper, all three models are evaluated—across three evaluation—across three evaluation sections: In-distribution (FF++, Domain A), Cross-generator (DFDC, Domain B), and Cross-population (Indian Face Set, Domain C). LoRA-r8 achieves better classification across all evaluation domains, outperforming both the bespoke CNN baseline and LNCLIP-DF as shown in [Table 4.1](#). All three models exhibit strong detection capabilities on the in-distribution test set (FF++), achieving high AUROCs as shown in [Figure 4.1](#). However, the bespoke CNN baseline performs worse on Cross-Generator Shift (DFDC) and Cross-Population Shift (Indian Set) when compared to LoRA-r8 and LNCLIP_DF.

Table 4.1 Performance of LoRA-r8, LNCLIP-DF, and the Xception across evaluation domains.

Model	AUROC _A	AUROC _B	AUROC _C	MCC _A	MCC _B	MCC _C
Xception	0.962	0.698	0.585	0.840	0.350	0.160
LNCLIP-DF	0.978	0.842	0.735	0.890	0.610	0.420
LoRA-r8	0.988	0.865	0.762	0.928	0.660	0.470

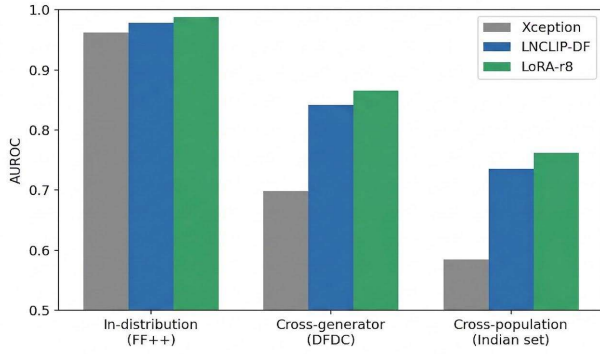


Figure 4.1 Illustrates the comparison of AUROC-scores of all three trained models on the three proposed test domains

To test whether high ranking scores actually translate to real world performance, **Figure 4.2** compares ranking metric - AUROC against threshold-dependant - Matthews Correlation Coefficient (MCC) evaluated at a fixed decision threshold calibrated strictly on the in-distribution validation split (FF++). The metric comparison exposes that decision thresholds decay when models face distribution shifts. In-distribution ranking accuracy translates directly into operational performance. Under domain shifts, in-distribution calibrated under decision thresholds become unreliable. Under cross-population shift, Xception's operational capacity severely degrades. PEFT models are not affected much by threshold decay and they do retain some operational utility under cross-population shift without any recalibration. Getting scores of 0.420 for LNCLIP-DF and 0.470 for LoRA-r8.

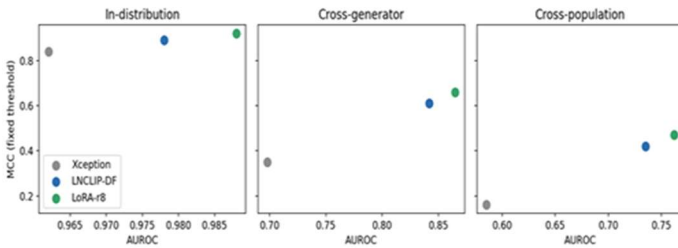


Figure 4.2 compares AUROC against MCC for all three models in different distribution shifts.

Figure 4.3 evaluates cross-population robustness across distinct face-swap algorithms within the Indian Face Set, breaking down the performance by its constituent generators: InSwapper and SimSwap. The per-generator breakdown demonstrates consistency across both synthesis pipelines. Across all three models, Inswapper outperforms others in terms of detection scores. LoRa-r8 yields superior performance as compared to LNCLIPDF and standard Xception across both sub-generators. This consistent trend proves that cross-population performance drops stem from broader demographic and domain shifts rather than flaws in any single generator.

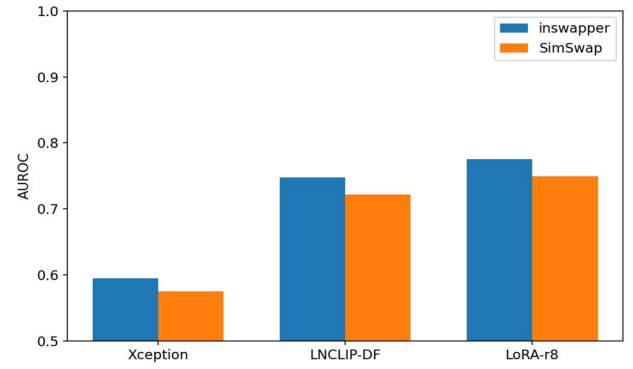


Figure 4.3 Illustrates the cross-population robustness of all three models across distinct face-swap algorithms

6. Conclusion

This work looks into whether cross-generator generalization of **parameter-efficient fine-tuning (PEFT)** on pretrained foundation models persists when tested on regional and demographic distribution shifts, or if its effectiveness reduces with bespoke CNN baselines. By decoupling generator architecture shift (DFDC) from compound demographic shift (IndicFairFace target set re-synthesized via inswapper and SimSwap), The study provided the first controlled empirical evaluation isolating these two axes of degradation. Sustained Performance Margins pose a strictly harder challenge than cross-generator shift for all models ($\Delta_{pop} > \Delta_{gen}$), PEFT methods do not experience a performance collapse relative to bespoke CNNs. Instead of shrinking or vanishing under demographic shift, PEFT's performance edge over fully fine-tuned CNN expands. On the cross-population test set, LoRA-r8 (AUROC = 0.762) and LNCLIP-DF (AUROC = 0.735) maintain substantial margins (+0.177 and +0.150 AUROC, respectively) over fully fine-tuned Xception (AUROC = 0.585), increasing their relative advantage over the CNN baseline compared to the cross-generator setting. With fixed operational decision thresholds set in distribution, PEFT architectures maintain effective classification ability (MCC = 0.420 -- 0.470), In contrast, the CNN baseline drops close to random guessing (MCC = 0.160).

These results show that adapting large pretrained foundation models via PEFT preserves rich, domain-invariant visual representations that resist performance reduction across both synthetic manipulation and human demographic variations. Updating less than 2% of model parameters (e.g., 0.18M in LNCLIP-DF or 0.42M in LoRA-r8) yields superior regional fairness and generalizability compared to updating tens of millions of parameters on task-specific CNNs (22.91M in Xception).

7. Future work

While the paper concluded that parameter-efficient foundation-model adaptation offers a robust framework,

several promising directions remain for future research. Demographic-aware PEFT: While zero-shot foundation models reduce demographic performance loss, there remains a notable gap compared to in-distribution baselines. Future efforts could explore fairness-oriented constraints into PEFT workflows. Expanding Backbone Architecture families: While CLIP ViT-B/16 has been tested, evaluating other architecture models like self-supervised vision models or multimodal LLMs to isolate the source of demographic robustness could be possible. Decoupling Demographic and Synthesis Variables: Focus on developing a generator-matched dataset to separate demographic shift from compound synthesis interactions. Dynamic Threshold Calibration: The MCC analysis study highlights the need for adaptive, zero-shot threshold adaptation mechanisms to ensure reliable deepfake detectors in real world use cases.

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